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Assessing Nitrogen Oxide Dispersion From a Natural Gas Compressor Station to Support Sustainable Energy Infrastructure Operation

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Abstract: Nitrogen oxide emissions from gas turbines at natural gas compressor stations pose a local environmental challenge to the sustainable operation of energy infrastructure. This study assesses nitrogen oxide dispersion around the Marand natural gas compressor station in northwestern Iran to identify candidate areas for ambient monitoring. Stack measurements from three natural gas-fired turbines were compiled to characterize nitric oxide (NO), nitrogen dioxide (NO_2), and total nitrogen oxides (NO_x). Meteorological data were processed using American Meteorological Society/Environmental Protection Agency Regulatory Model Improvement Committee Meteorological Preprocessor (AERMET), and dispersion was simulated using the American Meteorological Society/U.S. Environmental Protection Agency Regulatory Model (AERMOD). Maximum 1-hour and 24-hour concentrations were compared together with their receptor coordinates and occurrence times. The predicted maximum 1-hour concentrations of NO, NO_2, and NO_x were 90.50, 5.07, and 91.72 μg/m^3, respectively; the corresponding maximum 24-hour concentrations were 11.56, 0.88, and 11.71 μg/m^3. The hourly peak receptors were closely grouped, whereas the daily NO_2 and NO_x peak receptors were approximately 0.70 km from their respective hourly peak locations. The NO peak receptor remained unchanged between averaging periods. These results indicate that selecting monitoring locations from an hourly maximum alone may overlook locations associated with the highest daily concentrations. Joint interpretation of peak magnitude, location, and timing identifies candidate monitoring areas and provides information relevant to the environmental management and sustainable operation of natural gas transmission infrastructure.

Keywords: Natural gas compressor station; Gas turbine emissions; Nitrogen oxides; Atmospheric dispersion; AERMOD; Sustainable operation

1 Introduction

Natural gas compressor stations maintain the pressure required to transport gas through transmission networks. Where compressors are driven by gas turbines, this function also entails combustion-related emissions, including nitrogen oxides (NO_x). The environmental management of these facilities therefore extends beyond fuel consumption and equipment performance to the distribution of pollutants around the station. Nitrogen dioxide (NO_2) is particularly relevant to local air quality because of its effects on human health, as reflected in the World Health Organization air quality guidelines [1]. Identifying where turbine emissions produce elevated ambient concentrations is consequently an important component of environmentally responsible operation.

Reducing emissions at the combustor and assessing their consequences in the surrounding atmosphere address different parts of this problem. Combustion controls act on pollutant formation and release, whereas dispersion assessment examines the concentrations resulting from those releases under specified atmospheric conditions. Technical assessments of natural gas combustion turbines describe several approaches to NO_x control, with applicability depending on turbine design and operating conditions [2]. However, stack measurements alone cannot identify the locations of maximum ground-level concentrations. Exhaust temperature and velocity affect plume behavior, while atmospheric transport and turbulent mixing determine how pollutants reach surrounding receptors. A spatial assessment is therefore needed to connect source conditions with monitoring priorities outside the facility.

American Meteorological Society/U.S. Environmental Protection Agency Regulatory Model (AERMOD) is an established tool for examining this connection. Its formulation represents dispersion from industrial sources through

atmospheric boundary-layer characteristics and treatments of plume behavior and terrain [3]. Evidence from model comparisons supports its use while also demonstrating the importance of application conditions. Jittra et al. [4] evaluated AERMOD and California Puff Model using emissions from 292 industrial point sources and observations at ten monitoring sites in Thailand. AERMOD performed better overall for the pollutants assessed, including the upper end of the concentration distribution. This comparison supports the selection of AERMOD for industrial dispersion assessment, but its outcome does not imply equivalent predictive performance for every source configuration or meteorological setting.

Studies in Iran show how source measurements and dispersion modeling can be combined to investigate emissions from energy and process facilities. Hallaji et al. [5] measured pollutants from seven olefin-plant stacks and coupled AERMOD with meteorological fields from the Weather Research and Forecasting model. Their analysis related the orientation of concentration contours to prevailing winds, demonstrating that the spatial consequences of emissions cannot be inferred from stack concentrations alone. Mousavi et al. [6] examined CO, CO_2, and SO_2 dispersion from refinery flares under continuous and non-continuous operation. Although flare releases differ from turbine exhausts, that study illustrates why operating conditions must be considered when interpreting facility-related concentration patterns. Together, these applications establish a useful methodological basis without providing concentration estimates transferable directly to a natural gas compressor station.

More recent work has placed greater emphasis on the quality and interpretation of the emissions supplied to dispersion models. Rivero et al. [7] combined mobile miniature differential optical absorption spectroscopy (mini-differential optical absorption spectroscopy) measurements collected during seven campaigns between 2022 and 2024 with AERMOD simulations of the Tula industrial complex in Mexico. Their approach connected measured emission fluxes with the distribution of pollutants around nearby communities. Emerson and Khan [8] subsequently incorporated probabilistic emission inputs into an AERMOD screening framework, using a virtual pulp mill to examine the implications of emission variability. These studies address different source types and assessment objectives, but both show that a concentration estimate needs to be understood in relation to the emissions and conditions it represents. This is also relevant to compressor stations, where measurements from individual turbines provide the source-specific basis for assessing their combined atmospheric contribution.

The spatial interpretation of model outputs is equally important. A concentration at one receptor does not necessarily characterize the surrounding area, particularly for pollutants with strong local gradients. Fang et al. [9] demonstrated that spatial representativeness affected the evaluation of modeled NO_2 against ground observations. Their study concerned chemical transport models rather than station-scale plume modeling, but it identifies a broader assessment issue: the spatial support of an observation or prediction must match the question being addressed. For a compressor station, reporting only the largest concentration omits the receptor location that makes the result useful for monitoring.

Monitoring studies also show that sampling design affects the information obtained from an observation network. Blanco et al. [10] evaluated how the design of mobile monitoring campaigns influenced air pollution exposure models, emphasizing the relationship between sampling coverage and the resulting predictions. A station-level dispersion assessment serves a more focused purpose: it can identify candidate locations for examining the modeled contribution of a particular facility. For that purpose, both concentration magnitude and averaging period matter. The highest hourly concentration describes a short-duration peak, whereas a daily maximum reflects concentrations averaged over a longer interval. Their locations need not coincide, so a monitoring location selected from one statistic may not represent the other.

The practical question at the Marand natural gas compressor station is therefore not simply whether NO_x disperse beyond the stacks, but where the largest predicted concentrations occur and whether those locations remain consistent across pollutants and averaging periods. Addressing this question requires the source measurements, concentration maps, and numerical receptor outputs to be interpreted together. This approach gives the assessment a specific operational use: identifying the spatial features that a monitoring plan should examine.

This study assesses nitric oxide (NO), NO_2, and NO_x dispersion around the Marand natural gas compressor station in northwestern Iran using measurements from three natural gas-fired turbines and AERMOD simulations driven by meteorological data processed with American Meteorological Society/Environmental Protection Agency Regulatory Model Improvement Committee Meteorological Preprocessor (AERMET). The analysis compares maximum 1-hour and 24-hour concentrations, identifies their receptor locations and occurrence times, and examines whether the spatial focus of the predicted peaks changes with the averaging period. Its contribution is the joint interpretation of peak magnitude, location, and timing to inform ambient monitoring around the station. By connecting measured turbine emissions with specific monitoring priorities, the assessment supports the environmental management and sustainable operation of natural gas transmission infrastructure.

2 Materials and Methods

2.1 Study Site and Source Characteristics

The study examined the Marand natural gas compressor station in East Azerbaijan Province, northwestern Iran. The station maintains pressure within the natural gas transmission network. Three natural gas-fired turbines, designated C, D, and E, were included in the dispersion assessment. Their exhaust stacks constituted the emission sources considered in the analysis of nitrogen oxide concentrations around the facility.

The station was located at 38.517565° N, 45.357283° E. Each exhaust stack had an internal diameter of 2.00 m. Sampling points were positioned 10 m above the gas entry point, while the stack outlets were 14 m above the same reference. The exhaust conditions associated with the measurements are summarized in Table 1.

Table 1. Exhaust conditions of the three gas turbines

Parameter	Unit	Turbine C	Turbine D	Turbine E
Fuel	—	Natural gas	Natural gas	Natural gas
Stack internal diameter	m	2.00	2.00	2.00
Exhaust temperature	°C	375	396	387
Exhaust velocity	m/s	7.7	7.5	7.3
Reported volumetric flow rate	m^3/min	1450.7	141.3	1375.3
Atmospheric pressure	mbar	893	893	894

Note: The reported flow rate for turbine D is inconsistent with the listed stack diameter and exhaust velocity. Calculations using $Q = 60\pi D^2 v/4$, where D is the stack diameter in metres, v is the exhaust velocity in m/s, and Q is the volumetric flow rate in m^3/min, give approximately 1451.4, 1413.7, and 1376.0 m^3/min for turbines C, D, and E, respectively. These calculated values were used only for a consistency check and do not establish the flow rates entered into the dispersion model. "—" = not applicable.

2.2 Stack Measurements and Pollutant Data

Emission data were compiled from laboratory measurements of the three turbine exhausts. The measurement reports identified a Testo 340 flue gas analyzer and provided values for NO, NO_2, NO_x, SO_2, CO, CO_2, and O_2. The nitrogen oxide measurements formed the focus of the dispersion assessment, while the remaining components described the accompanying exhaust composition.

Reported exhaust gas concentrations are presented in Table 2. NO and NO_2 were examined individually, while NO_x represented the total nitrogen oxide quantity reported in the measurement dataset. NO_x was thus distinguished from its constituent species rather than interpreted as an additional independent chemical compound.

Table 2. Reported exhaust gas concentrations

Component	Unit	Turbine C	Turbine D	Turbine E
NO	ppm	58	47	44
NO_2	ppm	3	2	2
NO_x	ppm	61	49	46
O_2	%	17.90	17.59	18.72
CO_2	%	1.76	1.93	1.29
CO	ppm	0	0	0
SO_2	ppm	0	0	0

Note: Zero denotes a reported instrument reading and does not establish an absolute absence of the compound; NO = nitric oxide; NO_2 = Nitrogen dioxide; NO_x = nitrogen oxides; O_2 = oxygen; CO_2 = carbon dioxide; CO = carbon monoxide; SO_2 = sulfur dioxide.

The measurement dataset also contained oxygen-corrected mass concentrations at a reference oxygen content of 15% (Table 3). These values were presented separately from the measured volumetric concentrations in Table 2. They describe exhaust conditions and were not treated as ambient concentrations or used to infer pollutant composition at the peak receptors.

The reported gas concentrations and oxygen-corrected values were used to characterize the exhaust composition of the three turbines. These source measurements were distinguished from the receptor concentrations predicted by AERMOD. Comparisons among turbines were restricted to the measured exhaust characteristics; turbine-specific contributions to ambient concentrations were not inferred from stack concentrations alone.

2.3 Meteorological Data and Processing

Meteorological information for the station surroundings was obtained from Solcast. The variables included air temperature, relative humidity, wind speed, wind direction, and atmospheric pressure. Bright [11] evaluated Solcast's historical solar irradiance product; that validation does not establish the accuracy of the meteorological variables used here.

Table 3. Reported exhaust pollutant concentrations corrected to 15% reference oxygen

Component	Unit	Turbine C	Turbine D	Turbine E
NO	mg/Nm^3	153	112	159
NO_2	mg/Nm^3	12	7	11
NO_x	mg/Nm^3	165	120	171
CO	mg/Nm^3	0	0	0
SO_2	mg/Nm^3	0	0	0

Note: Nm^3 denotes a normalized gas volume. The corrected concentrations are presented on the measurement-report basis and are distinguished from the ambient concentrations predicted by American Meteorological Society/U.S. Environmental Protection Agency Regulatory Model (AERMOD); NO = nitric oxide; NO_2 = Nitrogen dioxide; NO_x = nitrogen oxides; O_2 = oxygen; CO = carbon monoxide; SO_2 = sulfur dioxide.

Meteorological data were processed using AERMET for the AERMOD simulations. AERMET produces surface and profile files that describe the atmospheric conditions used in dispersion calculations [12]. Wind direction, wind speed, and boundary-layer turbulence influence transport and dilution within this framework.

The present analysis examined the resulting receptor concentrations and peak locations. Separate sensitivity calculations were not conducted to quantify the effects of individual meteorological variables.

2.4 Atmospheric Dispersion Assessment

AERMOD predictions were evaluated to characterize the spatial distribution of NO, NO_2, and NO_x around the station. AERMOD uses a steady-state plume formulation based on atmospheric boundary-layer turbulence and scaling relationships [3, 13].

The analysis used source-group concentration summaries and receptor-specific contour maps. The numerical summaries provided ranked concentrations, projected receptor coordinates, and occurrence times. The contour maps were examined alongside these summaries to relate concentration magnitudes to their spatial distributions. Individual turbine contributions were not estimated.

NO, NO_2, and NO_x outputs were examined separately. The NO_2 values were interpreted as predictions conditional on the model configuration, rather than as an independently verified representation of total ambient NO_2. Differences among the outputs were not used to quantify atmospheric conversion between NO and NO_2.

Comparisons of concentration magnitude were made within a common averaging period. Independently selected pollutant maxima were not added because their receptor locations could differ. The contour maps represented the highest 1-hour concentration at each receptor over the modeled period, rather than a single plume occurring simultaneously throughout the domain.

2.5 Temporal and Spatial Comparison of Concentration Maxima

The analysis compared maximum 1-hour and 24-hour average concentrations for each pollutant. The hourly maximum characterizes a short-duration peak, whereas the daily maximum represents the highest concentration averaged over a 24-hour interval.

The contrast between these metrics was expressed as:

$$R_p = \frac{C_{p,\,\mathrm{max}}^{1\ \mathrm{h}}}{C_{p,\,\mathrm{max}}^{24\ \mathrm{h}}} \tag{1}$$

where, R_p is the dimensionless peak ratio for pollutant p, while $C_{p,\mathrm{max}}^{1\mathrm{h}}$ and $C_{p,\mathrm{max}}^{24\mathrm{h}}$ are its maximum 1-hour and 24-hour concentrations, respectively, expressed in µg/m^3. Ratios were calculated using unrounded model values. Because the maxima were selected independently, R_p describes their magnitude contrast rather than a temporal change at a fixed receptor.

The coordinates and occurrence times associated with each maximum were then compared. The straight-line separation between the hourly and daily peak receptors was calculated as:

$$D_p = \sqrt{\left(x_{p,1\ \mathrm{h}} - x_{p,24\ \mathrm{h}}\right)^2 + \left(y_{p,1\ \mathrm{h}} - y_{p,24\ \mathrm{h}}\right)^2} \tag{2}$$

where, D_p is the receptor separation in metres, and x and y are the projected coordinates associated with the model outputs. The coordinate displacement of the daily peak receptor relative to the hourly peak receptor was calculated as:

$$\Delta x_p = x_{p,24\text{ h}} - x_{p,1\text{ h}}, \Delta y_p = y_{p,24\text{ h}} - y_{p,1\text{ h}} \tag{3}$$

Positive values of Δx_p and Δy_p indicate increases in the projected x and y coordinates, respectively. These calculations describe differences between peak receptor locations; they do not represent plume travel distances.

Peak receptors were compared across pollutants and averaging periods to determine whether a common location represented all concentration maxima. This comparison provided the basis for identifying candidate areas for ambient monitoring around the station and assessing whether hourly and daily concentration patterns required attention at different locations.

3 Results

3.1 Predicted Concentrations Across Averaging Periods

The maximum predicted concentrations differed among pollutants and averaging periods (Table 4). The highest 1-hour concentrations of NO, NO_2, and NO_x were 90.50, 5.07, and 91.72 μg/m^3, respectively. The corresponding maximum 24-hour concentrations were 11.56, 0.88, and 11.71 μg/m^3. NO and NO_x had similar maximum concentrations within each averaging period, whereas the predicted NO_2 maxima were lower.

The ratios of the maximum 1-hour to maximum 24-hour concentrations, calculated using Eq. (1), were 7.83 for NO, 5.78 for NO_2, and 7.83 for NO_x. The hourly maxima therefore exceeded the daily maxima by different proportions across the pollutants. These ratios compared independently selected maxima and did not describe concentration changes at a fixed receptor.

Table 4. Maximum predicted 1-hour and 24-hour concentrations and their ratios

Pollutant	Maximum 1-Hour Concentration (μg/m^3)	Maximum 24-Hour Concentration (μg/m^3)	Ratio of Hourly to Daily Maxima
NO	90.50020	11.56477	7.83
NO_2	5.06525	0.87566	5.78
NO_x	91.71752	11.70981	7.83

Note: Ratios were calculated using unrounded model concentrations; NO = nitric oxide; NO_2 = Nitrogen dioxide; NO_x = nitrogen oxides.

3.2 Spatial Distribution of Maximum Hourly Concentrations

Figures 1–3 present the spatial distributions of the highest 1-hour concentrations of NO, NO_2, and NO_x. Each distribution contained a localized high-concentration zone surrounded by lower concentration contours. The contours were asymmetric, and the gradients differed around the high-value zones. The concentration field therefore could not be described as a uniform decline with distance from the station. Figures 1–3 retain the original concentration contours, with reformatted labels and coordinate axes approximately reconstructed from the original graphical outputs. Quantitative comparisons use the numerical model summaries.

The NO map showed a compact maximum within a broader area of lower concentrations (Figure 1). The NO_2 maximum occurred in the same general part of the modeled domain, although the associated concentration scale was substantially lower (Figure 2). The NO_x map also placed its highest concentration within this area (Figure 3). The three maps showed a broadly consistent spatial focus for hourly peaks, while the numerical summaries identified the corresponding receptor coordinates.

The maps used different concentration scales. Similar colors across the figures consequently did not indicate equal concentrations, and differences in mapped contour extent were interpreted with reference to the numerical legend for each pollutant. The highest 1-hour distributions also represented receptor-specific maxima over the modeled period rather than a single simultaneous plume. The timing of the overall maxima was therefore examined separately.

3.3 Locations and Timing of Hourly Peaks

The maximum 1-hour concentrations of all three pollutants occurred during the same reported hour on 19 March 2021 (Table 6). Their receptor locations were close but did not coincide. The NO maximum occurred at projected coordinates of $x = 531{,}619.33$ m and $y = 4{,}262{,}314.51$ m. The corresponding coordinates were 531,661.70 m and 4,262,290.80 m for NO_2, and 531,632.63 m and 4,262,304.48 m for NO_x.

Across the three hourly peak receptors, the projected x coordinates spanned 42.37 m and the y coordinates spanned 23.71 m. The hourly results therefore identified a closely grouped set of peak locations. The NO_2 peak

receptor had a higher x coordinate and a lower y coordinate than the NO peak receptor, while the NO_x peak receptor lay between them in both coordinate dimensions. These differences describe the reported peak coordinates. Their physical significance cannot be resolved without confirming that the pollutant simulations used comparable receptor grids and source configurations.

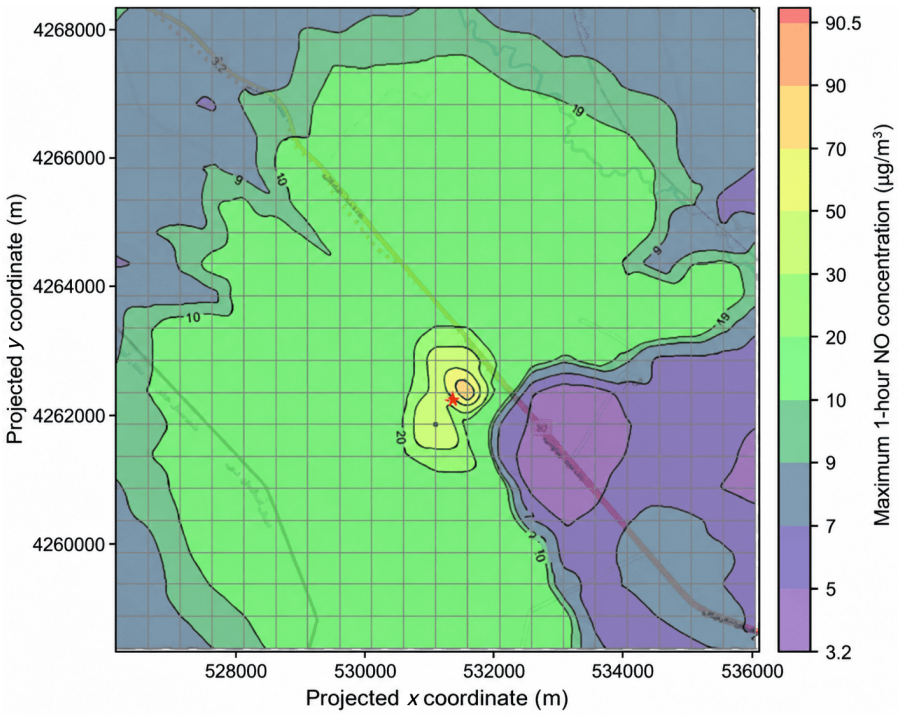


Figure 1. Maximum 1-hour NO concentration distribution

Note: NO = nitric oxide.

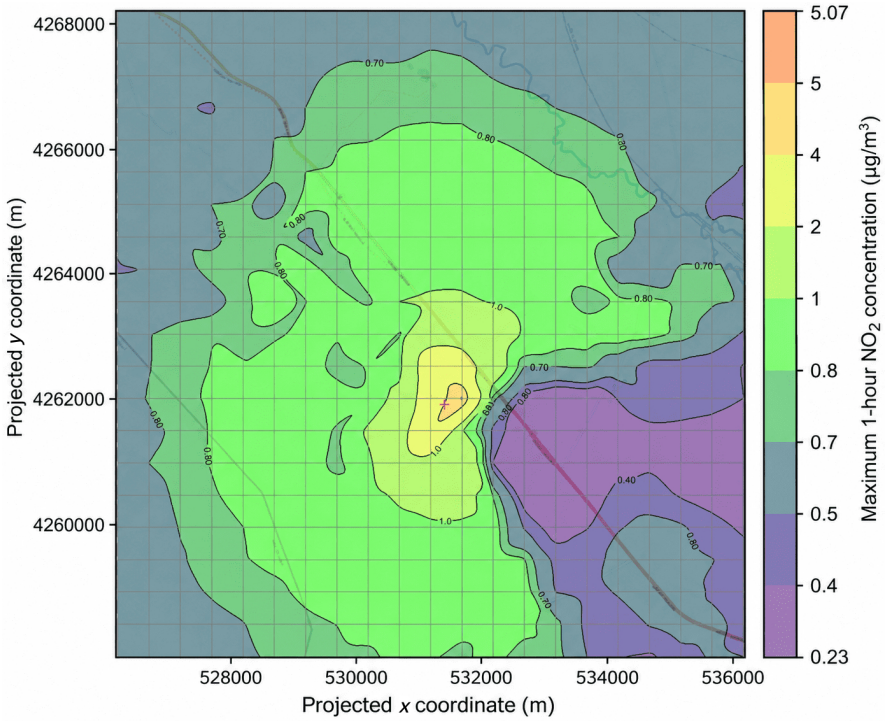


Figure 2. Maximum 1-hour NO_2 concentration distribution

Note: NO_2 = Nitrogen dioxide.

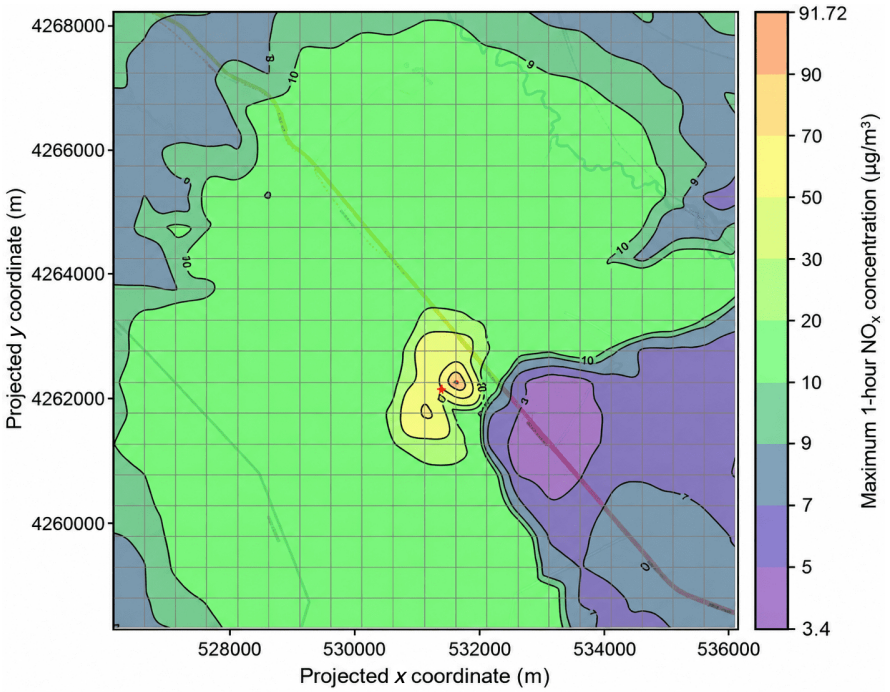


Figure 3. Maximum 1-hour NO_x concentration distribution

Note: NO_x = nitrogen oxides.

The shared peak hour established temporal coincidence for the three overall hourly maxima. Because the maxima were selected independently at different receptors, they were not used to infer pollutant composition at a common location. Accordingly, comparisons among the maxima were treated as comparisons of independently ranked concentration metrics.

3.4 Differences Between Hourly and Daily Peak Locations

The spatial arrangement of the daily maxima differed from that of the hourly maxima (Table 5). The maximum 24-hour NO concentration occurred at the same receptor as its hourly maximum. In contrast, the maximum 24-hour NO_2 and NO_x concentrations occurred at lower x coordinates and higher y coordinates than their respective hourly peak receptors.

Table 5. Locations and occurrence times of maximum 24-hour concentrations

Pollutant	Projected x Coordinate (m)	Projected y Coordinate (m)	Peak Date	Start-Hour Label
NO	531,619.33	4,262,314.51	18 March 2021	24
NO_2	531,163.41	4,262,787.95	17 February 2021	24
NO_x	531,134.91	4,262,800.49	17 February 2021	24

Note: Hour 24 retains the reporting convention of the model summaries; NO = nitric oxide; NO_2 = Nitrogen dioxide; NO_x = nitrogen oxides.

Table 6. Locations and occurrence times of maximum 1-hour concentrations

Pollutant	Projected x Coordinate (m)	Projected y Coordinate (m)	Peak Date	Start-Hour Label
NO	531,619.33	4,262,314.51	19 March 2021	10
NO_2	531,661.70	4,262,290.80	19 March 2021	10
NO_x	531,632.63	4,262,304.48	19 March 2021	10

Note: Coordinates are reproduced in the projected coordinate system of the model outputs; no geographic coordinate transformation was applied; NO = nitric oxide; NO_2 = Nitrogen dioxide; NO_x = nitrogen oxides.

For NO_2, the daily peak receptor had an x coordinate 498.29 m lower and a y coordinate 497.15 m higher than the

hourly peak receptor. The corresponding differences for NO_x were -497.72 m in x and +496.01 m in y. Applying Eq. (2) gave straight-line separations of 703.88 m for NO_2 and 702.67 m for NO_x (Table 7).

Table 7. Spatial offsets of daily peak receptors relative to hourly peak receptors

Pollutant	x-Coordinate Offset, Δx (m)	y-Coordinate Offset, Δy (m)	Receptor Separation (m)	Coordinate Displacement
NO	0.00	0.00	0.00	Same location
NO_2	-498.29	497.15	703.88	Lower x and higher y
NO_x	-497.72	496.01	702.67	Lower x and higher y

Note: Offsets and separations were calculated from the coordinates in Tables 5 and 6 using Eqs. (2) and (3). They describe differences between independently selected peak receptors; NO = nitric oxide; NO_2 = Nitrogen dioxide; NO_x = nitrogen oxides.

4 Discussion

4.1 Interpretation of Concentration Magnitudes and Spatial Patterns

The simulations identified localized concentration maxima rather than a uniform decline with distance from the station. This distinction is important for elevated turbine exhausts: the release point and the location of the highest ground-level concentration need not coincide. Plume rise and atmospheric mixing determine where an elevated release reaches ground-level receptors, while interactions with nearby structures can modify its dispersion. Research on AERMOD building downwash and plume rise has demonstrated the sensitivity of near-source predictions to these processes [14]. Evaluations of highly buoyant plumes have shown that the treatment of plume penetration and subsequent mixing into the convective boundary layer can affect predicted ground-level concentrations [15]. These mechanisms provide a physical context for the asymmetric patterns at Marand, although the present results do not isolate their individual contributions.

The similarity between the maximum NO and NO_x concentrations was consistent with the predominance of NO in the measured exhaust composition. However, this agreement should not be interpreted as a quantitative determination of ambient nitrogen oxide speciation. The independently ranked maxima occurred at different receptors, and the NO_x reporting basis must be considered when comparing mass concentrations. Under the modeled conditions, NO and NO_x had similar peak concentrations, while the NO_2 maxima were lower.

The lower NO_2 predictions also require a distinction between directly emitted NO_2 and NO_2 formed after release. Atmospheric conversion depends on chemical interactions, including the reaction of NO with ozone and the photolysis of NO_2. Carruthers et al. [16] showed that alternative chemical treatments within AERMOD produced different NO_2 predictions when evaluated against field datasets. Consequently, the comparatively low NO_2 values at Marand characterize the modeled concentration field; they do not establish that secondary NO_2 formation is unimportant.

4.2 Averaging Period and the Location of Maximum Concentrations

The comparison between averaging periods revealed information that was not apparent from concentration magnitude alone. Although the hourly peak receptors were closely grouped, the daily NO_2 and NO_x maxima occurred approximately 0.70 km from their respective hourly peak receptors. The daily NO maximum remained at its hourly peak receptor. Thus, spatial agreement among pollutants depended on the averaging period.

This difference is physically plausible because hourly and daily maxima represent different temporal selections. An hourly maximum can arise from a brief combination of transport and mixing conditions, whereas a daily maximum reflects concentrations averaged over a 24-hour interval. A receptor experiencing the largest single-hour concentration need not experience the largest daily average. The Marand outputs demonstrated this distinction directly through the coordinates associated with each maximum. They did not, however, establish which meteorological sequence produced the change in peak location.

The peak ratios provide a complementary description. Maximum hourly concentrations were approximately 7.83 times the daily maxima for NO and NO_x and 5.78 times the daily maximum for NO_2. These ratios summarize the contrast between two assessment metrics. They are not measures of concentration variability at a fixed location, nor do they indicate how frequently peak conditions occurred. Their practical value lies in showing that one averaging period cannot substitute for the other when describing the upper end of the predicted concentration field.

The three hourly maxima occurred during the same reported hour, whereas the daily maxima differed in occurrence date and receptor location. This combination shows why peak magnitude, position, and timing should be retained together. Reporting only the largest concentration would conceal whether different pollutants or averaging periods identify the same area for further investigation.

4.3 Comparison With Related Industrial and Compressor Station Studies

The Marand findings are consistent with the broader observation that industrial concentration patterns depend on source characteristics and atmospheric transport. Hallaji et al. [5] linked the orientation of modeled concentration contours around an Iranian olefin plant to prevailing winds. Rivero et al. [7] combined measured emissions with dispersion modeling to identify communities affected by an industrial complex in Mexico. The present assessment focused on three turbine stacks at a natural gas compressor station and compared the receptor locations associated with hourly and daily concentration maxima.

Direct numerical comparisons with those facilities would be misleading without harmonizing emission rates, stack conditions, receptor arrangements, meteorological periods, and pollutant treatment. A lower predicted concentration at Marand cannot by itself establish cleaner turbine operation or superior environmental performance. The useful comparison concerns the assessment approach: source measurements describe emissions, while dispersion outputs locate their potential atmospheric consequences.

Compressor station research provides a particularly relevant basis for interpreting model reliability. Panek et al. [17] evaluated AERMOD at a natural gas compressor station using concurrent emissions, meteorology, ambient NO_x, and ozone measurements collected over 13 months. Their analysis showed that assumptions about emissions and meteorological inputs could materially affect modeled-to-observed comparisons. Although that station used reciprocating engines rather than the gas turbines assessed here, the study demonstrates why general acceptance of a dispersion model does not establish its accuracy for an individual facility.

The contribution of the Marand analysis is site-specific: it identifies differences in peak magnitude, receptor location, and occurrence time between hourly and daily concentration metrics. These comparisons provide information for selecting candidate monitoring areas around the station.

4.4 Implications for Monitoring and Sustainable Operation

The separation between hourly and daily peak locations identifies candidate areas for subsequent monitoring assessment. A location selected solely from the hourly NO_x maximum would not coincide with the predicted daily NO_2 or NO_x maximum. The hourly and daily peak areas therefore warrant joint examination before selecting monitoring locations.

These modeled locations should guide an initial screening of monitoring options rather than determine an installation layout on their own. Final placement also depends on access, site boundaries, instrument exposure, nearby obstructions, and the monitoring objective. A monitoring site intended to investigate short-duration peaks may require a different placement strategy from one intended to characterize daily concentrations or background conditions.

This interpretation agrees with monitoring research showing that sampling design affects the information obtained from an observation network [10]. Liu et al. [18] further demonstrated, for coal-port operations, that accounting for uncertain operational and wind conditions could improve the robustness of monitoring layouts. Their optimization method was not applied in the present study, but it supports the principle that monitoring decisions should consider more than one concentration scenario.

For Marand, the results support consideration of both peak areas, potentially through an initial mobile campaign or a combination of fixed and temporary measurements. Such measurements could establish whether the modeled spatial distinction persists under actual operating conditions.

The relevance to sustainable operation lies in directing environmental observations toward locations where elevated station-related concentrations are predicted. Measurements at these locations could establish a baseline for evaluating subsequent operational or emission-control changes.

4.5 Uncertainty and Applicability

The predictions are conditional on the source characteristics, meteorological inputs, and dispersion assumptions. Stack measurements describe the sampled operating conditions, whereas turbine utilization and emissions may vary over time. Emerson's probabilistic assessment [8] illustrates how emission variability can be incorporated into AERMOD screening. The Marand results should therefore be interpreted as concentration patterns associated with the assessed conditions rather than as a complete description of all operating states.

The assessment focuses on the interpretation of the concentration outputs. Full reproduction of the dispersion calculations requires source-specific mass emission rates, concentration conversion conventions, the meteorological processing configuration, receptor-grid specifications, and the treatment of NO-to-NO_2 conversion. These elements are not fully documented in the present study. The results therefore support comparisons within the evaluated model outputs, while absolute concentrations and monitoring locations require further verification before operational or regulatory application.

The representation of plume behavior and meteorology also affects predictive performance. Pandey et al. [19] found that incorporating aircraft plume rise and shoreline meteorology improved agreement between AERMOD

predictions and observations around Los Angeles airport. Although those conditions differ from the Marand compressor station, their evaluation demonstrates the importance of matching model treatments to the physical setting. The asymmetric concentration patterns at Marand are consistent with spatially varying dispersion, but the present analysis does not isolate the effects of plume rise, atmospheric stability, or surrounding structures.

Interpretation also depends on the averaging period and assessment objective. Tan et al. [20] examined industrial contributions using hourly, daily, and annual AERMOD outputs, illustrating the different temporal scales relevant to facility-related air pollution. At Marand, the separation between hourly and daily peak receptors similarly shows why a single concentration statistic is insufficient for identifying monitoring priorities. Wu et al. [21] extended AERMOD simulations around liquefied natural gas terminals to health and equity assessment by combining concentration estimates with additional population and health information. Such applications demonstrate the wider relevance of near-source modeling, while also clarifying that the present concentration results alone do not establish population exposure or health effects.

Field observations are needed to evaluate whether the predicted high-concentration areas correspond to actual station-related patterns. Dehkhoda et al. [22] combined remote and in situ measurements with Hybrid Single-Particle Lagrangian Integrated Trajectory (HYSPLIT) simulations to examine industrial emissions and their transport toward surrounding areas. Their approach illustrates how complementary measurements can be used to investigate industrial emission patterns and their downwind influence. For Marand, concurrent nitrogen oxide, meteorological, and turbine operating measurements would allow the modeled hourly and daily peak areas to be examined under comparable conditions.

Monitoring around the station would require calibration and reference comparisons, particularly if compact sensor networks were used to extend spatial coverage. Studies of NO and NO_2 sensors have demonstrated the importance of deployment conditions, ozone interference, and scalable calibration procedures [23, 24]. Aleahmad et al. [25] combined AERMOD predictions with the analytic hierarchy process to select monitoring locations in an Iranian petrochemical zone. Their approach illustrates how dispersion outputs can be integrated with additional siting criteria; the peak locations identified at Marand represent an initial screening step rather than an optimized monitoring network.

Finally, peak coordinates identify the highest values among the receptors evaluated and do not establish the exact location of a continuous atmospheric maximum. NO_2 interpretation additionally depends on the treatment of atmospheric conversion [16], while ambient compliance assessment requires background concentrations and contributions from other sources. Within these boundaries, the results provide candidate monitoring areas and a basis for testing the station's local air quality influence.

5 Conclusions

This study assessed nitrogen oxide dispersion around the Marand natural gas compressor station using stack measurements from three gas turbines and AERMOD simulations. The analysis compared predicted concentration magnitudes, peak receptor locations, and occurrence times to identify candidate areas for ambient monitoring.

The maximum 1-hour concentrations of NO, NO_2, and NO_x were 90.50, 5.07, and 91.72 $\mu g/m^3$, respectively, while the corresponding maximum 24-hour concentrations were 11.56, 0.88, and 11.71 $\mu g/m^3$. The ratios of hourly to daily maxima were 7.83, 5.78, and 7.83, demonstrating the different concentration magnitudes represented by these averaging periods. The dispersion maps showed localized, asymmetric high-concentration zones rather than a uniform decrease with distance from the station.

The hourly maxima occurred at closely grouped receptors during the same reported hour. However, the daily NO_2 and NO_x maxima occurred approximately 704 and 703 m from their respective hourly peak receptors, whereas the NO peak location remained unchanged. A location selected solely from an hourly maximum therefore did not coincide with every daily maximum.

The principal contribution is the joint interpretation of peak magnitude, location, and timing as a basis for monitoring decisions. For Marand, the hourly peak cluster and the separate daily NO_2 and NO_x peak cluster identify candidate areas for further monitoring assessment. Their operational use requires confirmation of spatial resolution and comparison with ambient observations.

The findings describe the modeled station contribution under the assessed conditions. Their application to ambient air quality compliance requires consideration of background concentrations, NO-to-NO_2 conversion, and comparison with concurrent ambient observations.

Author Contributions

Conceptualization, M.M.; methodology, A.R.; software, A.R.; formal analysis, A.R.; investigation, N.R. and A.R.; resources, N.R.; data curation, A.R.; writing—original draft preparation, N.R.; writing—review and editing, M.M. and A.A.; visualization, N.R.; supervision, M.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] World Health Organization, "WHO global air quality guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide," 2021. https://www.who.int/publications/i/item/9789240034228

[2] Sargent & Lundy, "Combustion turbine NO_x control technology memo," Eastern Research Group, Inc., Chicago, IL, USA, Project No. 13527-002, 2022. https://www.epa.gov/system/files/documents/2023-12/attachment-5-10-combustion-turbine-NOX-control-technology-memo.pdf

[3] A. J. Cimorelli, S. G. Perry, A. Venkatram, J. C. Weil, R. J. Paine, R. B. Wilson, R. F. Lee, W. D. Peters, and R. W. Brode, "AERMOD: A dispersion model for industrial source applications. Part I: General model formulation and boundary layer characterization," *J. Appl. Meteorol. Climatol.*, vol. 44, pp. 682–693, 2005. https://doi.org/10.1175/jam2227.1

[4] N. Jittra, N. Pinthong, and S. Thepanondh, "Performance evaluation of AERMOD and CALPUFF air dispersion models in industrial complex area," *Air Soil Water Res.*, vol. 8, p. ASWR.S32781, 2015. https://doi.org/10.4137/aswr.s32781

[5] H. Hallaji, M. R. Bohloul, S. M. Peyghambarzadeh, and S. Azizi, "Measurement of air pollutants concentrations from stacks of petrochemical company and dispersion modeling by AERMOD coupled with WRF model," *Int. J. Environ. Sci. Technol.*, vol. 20, pp. 7217–7236, 2023. https://doi.org/10.1007/s13762-023-04959-w

[6] S. S. Mousavi, G. Goudarzi, S. Sabzalipour, M. M. Rouzbahani, and E. M. Hassan, "An evaluation of CO, CO_2, and SO_2 emissions during continuous and non-continuous operation in a gas refinery using the AERMOD," *Environ. Sci. Pollut. Res.*, vol. 28, pp. 56 996–57 008, 2021. https://doi.org/10.1007/s11356-021-14493-2

[7] R. A. G. Rivero, C. I. R. Cárdenas, H. A. B. Huertas, and M. T. Vázquez, "Assessment of NO_2 and SO_2 emissions in the Tula industrial complex, Hidalgo, using the mobile mini-DOAS technique combined with the AERMOD dispersion model," *Environ. Monit. Assess.*, vol. 197, pp. 1–20, 2025. https://doi.org/10.1007/s10661-025-14265-2

[8] Z. I Emerson and T. R. Khan, "Beyond deterministic air quality modeling: A probabilistic screening approach for emission inputs in AERMOD," *Integr. Environ. Assess. Manag.*, vol. 21, pp. 1358–1367, 2025. https://doi.org/10.1093/inteam/vjaf098

[9] L. Fang, J. Jin, A. Segers, K. Li, J. Xia, W. Han, B. Li, H. X. Lin, L. Zhu, S. Liu *et al.*, "Observational operator for fair model evaluation with ground NO_2 measurements," *Geosci. Model Dev.*, vol. 17, pp. 8267–8282, 2024. https://doi.org/10.5194/gmd-17-8267-2024

[10] M. N. Blanco, J. Bi, E. Austin, T. V. Larson, J. D. Marshall, and L. Sheppard, "Impact of mobile monitoring network design on air pollution exposure assessment models," *Environ. Sci. Technol.*, vol. 57, pp. 440–450, 2022. https://doi.org/10.1021/acs.est.2c05338

[11] J. M. Bright, "Solcast: Validation of a satellite-derived solar irradiance dataset," *Sol. Energy*, vol. 189, pp. 435–449, 2019. https://doi.org/10.1016/j.solener.2019.07.086

[12] United States Environmental Protection Agency, "User's guide for the AERMOD meteorological preprocessor (AERMET)," United States Environmental Protection Agency, EPA-454/B-26-005, 2026. https://gaftp.epa.gov/Air/aqmg/SCRAM/models/met/aermet/aermet_userguide.pdf

[13] United States Environmental Protection Agency, "User's guide for the AMS/EPA regulatory model (AERMOD)," United States Environmental Protection Agency, EPA-454/B-26-001, 2026. https://gaftp.epa.gov/Air/aqmg/SCRAM/models/preferred/aermod/aermod_userguide.pdf

[14] R. L. Petersen, J. O. Paumier, and S. A. Guerra, "Development, evaluation, and implementation of building downwash and plume rise enhancements in AERMOD," *J. Air Waste Manag. Assoc.*, vol. 72, pp. 1423–1441, 2022. https://doi.org/10.1080/10962247.2022.2120563

[15] C. J. Warren, R. J. Paine, J. A. Connors, C. Szembek, and E. Knipping, "Evaluation of a revised AERMOD treatment of plume dispersion in the daytime elevated stable layer," *J. Air Waste Manag. Assoc.*, vol. 72, pp. 1040–1052, 2022. https://doi.org/10.1080/10962247.2022.2094031

[16] D. J. Carruthers, J. R. Stocker, A. Ellis, M. D. Seaton, and S. E. Smith, "Evaluation of an explicit NO_x chemistry

method in AERMOD," *J. Air Waste Manag. Assoc.*, vol. 67, pp. 702–712, 2017. https://doi.org/10.1080/10962247.2017.1280096

[17] J. A. Panek, J. M. McCarthy, A. Z. Huth, A. J. Krol, and C. Nowak, "PRCI ambient NO_2 AERMOD performance assessment and model improvement project: Modeled to observed comparison," *J. Air Waste Manag. Assoc.*, vol. 70, pp. 504–521, 2020. https://doi.org/10.1080/10962247.2020.1743382

[18] B. Liu, Y. Peng, W. Wang, and N. Mao, "Robust optimization for designing air quality monitoring network in coal ports under uncertainty," *Atmos. Environ.*, vol. 304, p. 119792, 2023. https://doi.org/10.1016/j.atmosenv.2023.119792

[19] G. Pandey, A. Venkatram, and S. Arunachalam, "Accounting for plume rise of aircraft emissions and shoreline meteorology enhances AERMOD's description of concentrations measured around Los Angeles airport," *J. Air Waste Manag. Assoc.*, vol. 74, pp. 726–742, 2024. https://doi.org/10.1080/10962247.2024.2394104

[20] A. Z. L. Tan, W. S. Ho, M. H. Hassim, F. Abdullah, and L. Y. Lim, "Evaluation of industrial air pollution contribution by type of industry in Pasir Gudang using AERMOD," *Environ. Monit. Assess.*, vol. 197, pp. 1–24, 2025. https://doi.org/10.1007/s10661-025-13797-x

[21] X. Wu, T. Holloway, D. Amin, P. Meier, V. S. Limaye, and A. Samuel, "Assessing near-source health and equity impacts of liquefied natural gas terminals," *GeoHealth*, vol. 10, p. e2025GH001609, 2026. https://doi.org/10.1029/2025gh001609

[22] N. Dehkhoda, J. Sim, J. Shin, S. Joo, S. H. Cho, J. H. Kim, and Y. Noh, "Air pollution measurement and dispersion simulation using remote and in situ monitoring technologies in an industrial complex in Busan, South Korea," *Sensors*, vol. 24, p. 7836, 2024. https://doi.org/10.3390/s24237836

[23] A. Abu-Hani, J. Chen, V. Balamurugan, A. Wenzel, and A. Bigi, "Transferability of machine-learning-based global calibration models for NO_2 and NO low-cost sensors," *Atmos. Meas. Tech.*, vol. 17, pp. 3917–3931, 2024. https://doi.org/10.5194/amt-17-3917-2024

[24] A. R. Winter, Y. Zhu, N. G. Asimow, M. Y. Patel, and R. C. Cohen, "A scalable calibration method for enhanced accuracy in dense air quality monitoring networks," *Environ. Sci. Technol.*, vol. 59, pp. 2599–2610, 2025. https://doi.org/10.1021/acs.est.4c08855

[25] M. Aleahmad, A. Karbassi, A. H. Davami, and R. J. Yengejeh, "Air pollution modeling for the optimal location of air quality monitoring stations in the petrochemical special economic zone in Mahshahr (Iran)," *Environ. Model. Assess.*, vol. 31, pp. 261–272, 2025. https://doi.org/10.1007/s10666-025-10063-9